

RESEARCH OVERVIEW

My lab’s research brings a data-driven approach to understanding and improving the Internet’s performance and security, both on the ground and in space. We build systems that collect data about network, operator, and attacker behaviors. We analyze the data our systems collect to surface operational challenges and threats.

POSITIONS

- Assistant Professor, Electrical and Computer Engineering, *University of California, Los Angeles* July 2024–Present
- Research Scientist, *Netflix* June 2023–May 2024, May 2025–June 2026
- Research Scientist, *Censys* June 2024–May 2025

EDUCATION

- Ph.D. in Computer Science, *Stanford University* 2018–2024
Dissertation: “A Comprehensive and Real-Time View of the Internet Service Ecosystem”
Advisor: Zakir Durumeric
- M.S. in Computer Science, *University of California, San Diego* 2017–2018
Thesis: “Building and Breaking Burst-Parallel Systems”
Advisors: Geoff Voelker and George Porter
- B.S. in Computer Science, *University of California, San Diego* 2014–2017
Minor: Mathematics

HONORS AND AWARDS

- Hellman Fellow, UCLA 2026
- Forbes 30 Under 30, Science 2025
- Pulse Research Fellow (Mentor), Internet Society 2025
- EECS Rising Star, International Recognition 2023
- Community Impact Award, Stanford University 2023
- Student Services Award, Stanford University Computer Science Department 2023
- Community Contribution Paper Award, ACM Internet Measurement Conference 2022
- Graduate Research Fellowship, National Science Foundation (“NSF GRFP”) 2018
- Graduate Fellowship in Science and Engineering, Stanford University (“Stanford SGF”) 2018
- Department Award for Excellence in Teaching, UC San Diego 2018

Conference Proceedings

- [1] H. Clark, J. Cody, Z. Durumeric, M. Ellison, L. **Izhikevich**, and A. Mirian, “Censys: A Map of Internet Hosts and Services”, in *Proceedings of the ACM SIGCOMM Conference*, 2025.
- [2] M. Tran, K. Huynh, D. Jain, D. Truong, S. Theerananachai, B. Zhang, L. Zhang, and L. **Izhikevich**, “Towards Global Outage Detection for LEO Networks”, in *Proceedings of the ACM SIGCOMM LEO-Net Workshop*, 2025.
- [3] L. **Izhikevich**, R. Engardt, T. Huang, and R. Teixeira, “A Global Perspective on the Past, Present, and Future of Video Streaming over Starlink”, in *ACM SIGMETRICS/IFIP Performance*, 2025.
- [4] K. Izhikevich, G. Voelker, S. Savage, and L. **Izhikevich**, “Using Honeybuckets to Characterize Serverless Storage Scanning in the Wild”, in *IEEE European Symposium on Security and Privacy*, 2024.
- [5] L. **Izhikevich**, M. Tran, K. Izhikevich, G. Akiwate, and Z. Durumeric, “Democratizing LEO Satellite Network Measurement”, in *ACM SIGMETRICS/IFIP Performance*, 2024.
- [6] L. **Izhikevich**, M. Tran, M. Kallitsis, A. Fass, and Z. Durumeric, “Cloud Watching: Understanding Attacks Against Cloud-Hosted Services”, in *Proceedings of the 23rd ACM Internet Measurement Conference*, 2023.
- [7] L. **Izhikevich**, G. Akiwate, B. Berger, S. Drakontaidis, A. Ascherman, P. Pearce, D. Adrian, and Z. Durumeric, “ZDNS: A Fast DNS Toolkit for Internet Measurement”, in *Proceedings of the 22nd ACM Internet Measurement Conference*, **★Community Contribution Award★**, 2022.
- [8] L. **Izhikevich**, R. Teixeira, and Z. Durumeric, “Predicting IPv4 Services Across All Ports”, in *Proceedings of the ACM SIGCOMM Conference*, 2022.
- [9] M. Ziv, L. **Izhikevich**, K. Ruth, K. Izhikevich, and Z. Durumeric, “ASdb: A System for Classifying Owners of Autonomous Systems”, in *Proceedings of the 21st ACM Internet Measurement Conference*, 2021.
- [10] J. Cable, D. Gregory, L. **Izhikevich**, and Z. Durumeric, “Stratosphere: Finding Vulnerable Cloud Storage Buckets”, in *Proceedings of the 24th International Symposium on Research in Attacks, Intrusions and Defenses*, 2021.
- [11] L. **Izhikevich**, R. Teixeira, and Z. Durumeric, “LZR: Identifying Unexpected Internet Services”, in *30th USENIX Security Symposium*, 2021.
- [12] G. Wan, L. **Izhikevich**, D. Adrian, K. Yoshioka, R. Holz, C. Rossow, and Z. Durumeric, “On the Origin of Scanning: The Impact of Location on Internet-Wide Scans”, in *ACM Internet Measurement Conference*, 2020.
- [13] L. Ao, L. **Izhikevich**, G. M. Voelker, and G. Porter, “Sprocket: A Serverless Video Processing Framework”, in *Proceedings of the Ninth ACM Symposium on Cloud Computing*, 2018.
- [14] L. **Izhikevich**, E. Peterson, and B. Voytek, “Neural oscillatory power is not Gaussian distributed across time”, in *Program No. 271.03. 2016 Neuroscience Meeting Planner*, 2016.

Workshop/Posters/Pre-Prints

- [16] O. Elamri, I.-N. Zanoria, J. Zhi, B. Du, and L. **Izhikevich**, “Expanding Access, Exposing Risk: A Study of Starlink Hosts”, in *ACM Workshop on Policy-Relevant Internet Measurements and Experimentation (PRIME)*, 2025.
- [17] F. Steurer, A. Phokeer, P. Paeps, and **Izhikevich, Liz**, “Measuring resilience of authoritative DNS”, *Proceedings of the ACM SIGCOMM Posters and Demos*, 2025.

- [18] L. **Izhikevich**, R. Gao, E. Peterson, and B. Voytek, “Measuring the average power of neural oscillations”, *bioRxiv*, 2018. eprint: <https://www.biorxiv.org/content/early/2018/10/13/441626.full.pdf>.

Books

- [15] N. Moshiri and L. **Izhikevich**, *Design and Analysis of Data Structures*. 2016, ISBN: 978-1981017232.

Thesis

- [19] L. **Izhikevich**, “Building and Breaking Burst-Parallel Systems”, M.S. thesis, University of California, San Diego, 2018.
- [20] L. **Izhikevich**, “A Comprehensive and Real-Time View of the Internet Service Ecosystem”, Ph.D. dissertation, Stanford University, 2024.

PROFESSIONAL SERVICE

Technical Program Committees

• USENIX Security	2025–Current
• Internet Measurement Conference	2023–Current
• SIGCOMM Next-Generation Network Observability Workshop	2025
• IEEE Security and Privacy	2023
• Symposium on Research in Attacks, Intrusions, and Defenses	2022–2023
• The Passive and Active Measurement Conference	2022
• IEEE Security and Privacy (External Reviewer)	2022
• USENIX Security (External Reviewer)	2019–2022
• Internet Measurement Conference (External Reviewer)	2019–2021

Department and University Service

• Area Director for Systems and Signals ECE, UCLA	2026
• Institute for Digital Research and Education Board, UCLA	2026–2029
• DGE Eugene Cota-Robles Faculty Review Committee, UCLA	2026
• One-IT Working Group, UCLA	2026
• Computer Engineering Faculty Search Committee – Chalk Talk Coordinator, UCLA	2025
• Founder and Organizer of the Cross-Department Security Seminar, UCLA	2024–2025
• Data Science Faculty Search Committee, Stanford University	2022–2023
• Ethics & Society Review of HAI Seed Grants Committee, Stanford University	2022
• Chair of Ph.D. Applicant Support Program, Stanford University	2021–2023
• Ph.D. Admissions Committee, Computer Science, Stanford University	2019–2022

External Service

• Grant Proposal Reviewer, Dutch Research Council	2025
• Mentor for Pulse Research Fellows, Internet Society	2025

TEACHING

- ECE239AS: Advanced Networking and Security, UCLA Spring 2025, Spring 2026, Spring 2027
- ECE132B: Data Communications and Telecommunication Networks, UCLA Fall 2025, Winter 2026
- ECE133A: Applied Numerical Computing, UCLA Fall 2026

ADVISING AND MENTORING

Ph.D. Students

- Manda Tran (M.S., Ph.D.) 2024–Current
- Rahul Tripathi (M.S., Ph.D.) 2025–Current
- David Budziwojzki (M.S., Ph.D.) 2026–Current

Post-Docs

- Ben Du 2025–Current

B.S/M.S Students (only those with a peer-reviewed article)

- Khiet Hyunh (B.S), 2025–Current
- Omar Elamri (M.S), Isaac-Neil Zanolari (M.S), Jacob Zhi (M.S), Dravya Jain (B.S), Dylan Truong (B.S) 2025
- Briana Berger (B.S/M.S.), Spencer Drakontaidis (B.S.), Anna Ascherman (B.S) 2021–2022
- Jack Cable (B.S.), Drew Gregory (B.S.), Maya Ziv (M.S.) 2020–2021

Doctoral Committees

- Mine Dogan (Ph.D., Electrical and Computer Engineering, UCLA, May 2025)
- Howard Zhang (Ph.D., Electrical and Computer Engineering, UCLA, November 2025)
- Alexander Vilesov, (Ph.D., Electrical and Computer Engineering, UCLA, June 2026)
- Fatemeh Arkannezhad (Ph.D., Electrical and Computer Engineering, UCLA, TBD)

Masters Committees

- Jacob Zhi (M.S, Computer Science, UCLA, June 2025)
- Fatemeh Arkannezhad (M.S, Electrical and Computer Engineering, UCLA, June 2025)

SPEAKING

- Enabling Live Video Contribution Using LEO Satellites
Invited at IETF Space RG July'26
- Arctic 2040: National Security Blueprint for the Navy
Invited at Office of Naval Research National Security and Naval Power Bootcamp June'26
- Global Visibility into the Performance, Operations, and Security of LEO Networks
Invited at CMU CyLab Seminar Feb'26; Brown Systems Seminar Feb'26
- Global Visibility into the Internet's Security and Performance
Invited at UCLA Engineering Alumni Board Meeting May'25; UCLA ECE Alumni Board Meeting Nov'25
- Past, Present, and Future of Low Earth Orbit Satellite Networks
Invited at Aerospace Corporation May'25; Stanford University March'25; Northrop Grumman Nov'25
- How to Succeed at Early Career Research
Invited Panelist at the Internet Measurement Conference October'24; Invited Panelist on the Networking Channel March '25

- Manipulating Congestion Control To Improve Video Streaming Over Starlink
Invited at IETF July '25
- A Global Perspective on the Past, Present, and Future of Video Streaming over Starlink
Invited at Netflix May'24; ACM SIGMETRICS June '25;
- Democratizing LEO Satellite Network Measurement
Invited at Netflix June'23; ACM SIGMETRICS June'24
- How to give an interesting talk for a SIGCOMM/NSDI audience
Invited Panelist on the Networking Channel March '23
- Cloud Watching: Understanding Attacks Against Cloud-Hosted Services
ACM Internet Measurement Conference October'22
- ZDNS: A Fast DNS Toolkit for Internet Measurement
ACM Internet Measurement Conference October'22
- Predicting IPv4 Services Across All Ports
ACM SIGCOMM August'22
- Finding Vulnerable Cloud Storage Buckets
Symposium on Research in Attacks, Intrusions, and Defenses October'21; Invited for UC San Diego Security Seminar '22
- Identifying Unexpected Internet Services
USENIX Security '21; Invited for Cornell Tech Security Seminar '21; Invited for Stanford Security Symposium'22
- Sprocket: A Serverless Video Processing Framework
UC San Diego Systems Seminar '18

REFERENCES

Zakir Durumeric

Assistant Professor of Computer Science
Stanford University

Stefan Savage

Professor of Computer Science and Engineering
University of California, San Diego

Geoffrey Voelker

Professor of Computer Science and Engineering
University of California, San Diego

Renata Cruz Teixeira

Former Director of Research; Current Senior Research Scientist
Inria, Paris; Netflix